



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

A Network of Elements Within a System



Learning Outcome:

Understands systems theories, business processes and basic concepts on IT.

Create a map of processes that allows identifying areas in an information system, using IT tools to generate reports based on the company's needs.

Can Do Statements:

- I can talk about the concept of information.
- I can explain briefly what data is.
- I can talk about the concept of a system.

Ask your partner:

- How do you think the Internet classifies information?
- Why do you think the Internet doesn't show certain information??





Let's Explore!

Susan studies the main concepts of systems theory.



Instructions: Read the text below.

Alan: Good Morning Susan, the tutorial we have for this week is about the key elements for understanding computers, right?

Susan: Yes. There are some basic concepts that we need to understand when talking about computers. These are information, data and systems. Let's begin by talking about data!

Alan: Yes. We can understand this term as a group of facts that are not organized. These sometimes come in binary code, or series of 0s and 1s; and can seem useless or random as they do not have a context. We need to process and classify these facts before we can understand or use them.

Susan: Then, we have information. This refers to processed data.

Alan: Exactly. When we receive data, we enter it for the computer to process it and with simple formulas, the computer can transform data into information.

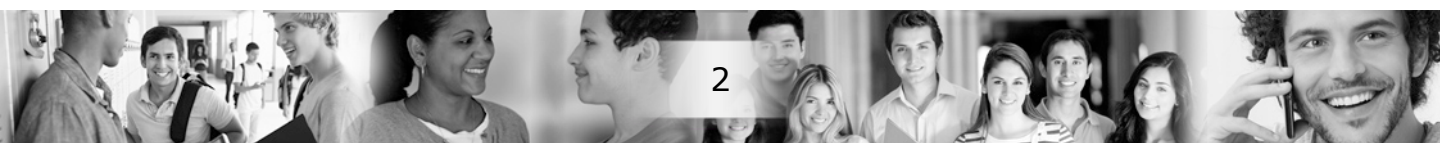
Susan: So basically, we can use data as a source of facts that the computer processes and shapes and then, we receive this as information. Once we have this information organized and processed, we can use it to make decisions or to run programs.

Alan: Now, what about systems?

Susan: This one is a little bit more complex. A system is a group of components that you organize for a common purpose. When we think of a system, we have to consider two things: the elements that compose it and their functions. This means that those elements or components have different functions or tasks inside the system and they work together to keep this system going.

Alan: I think it's all clear now, thank you so much for your help!

Susan: No problem! We can talk more about that tutorial whenever you want.





Let's Practice!

I understand that Information *turns into** data.

*When something "*turns into*" it changes or transforms.

Instructions: Complete each sentence using the words in the Word Bank.

Word Bank

Those / This / That

- _____ information concept in that book is related to data concept.
- _____ pieces of information turn into data with a simple formula.
- _____ information that appears here can transform into data after the computer processes it.



Let's Practice!

Let's keep talking about data!

Instructions: Complete the sentences using the hints.

- Data can become _____ when it reveals patterns about a given phenomenon. (HINT: i n _ _ _ _ _ _ _ _)
- We enter data in a computer for _____ it. (HINT: p r _ _ _ _ _ _ _ _)
- In some computers, data is in _____ code. (HINT: b _ _ _ _ y)



Extension Activity

Let's talk about information, data and systems.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about the terms: information, data and systems.
- Make a descriptive chart about these concepts.
- Write how these concepts work together and their importance.
- Share your descriptive chart with the class and write more characteristics

Self-Assessment

Yes

No

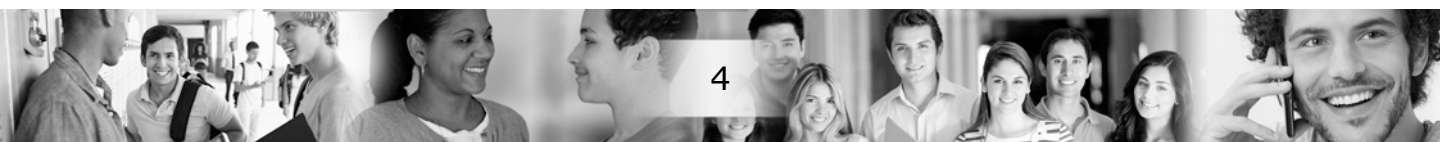
- I find information about the terms: information, data and systems concepts.
- I make organize information, data and systems concepts.
- I identify information, data and systems concepts importance.
- I share my descriptive chart with the class and write more characteristics.

☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- **binary** (adjective): consisting of two parts or has a combination with multiples of two
e.g. We can understand this term as a group of facts that are not organized. These sometimes come in binary code, or series of 0s and 1s.
- **random** (adjective): occurring for no particular reason or without a particular pattern, accidentally
e.g. These sometimes come in binary code, or series of 0s and 1s; and can seem useless or random as they do not have a context.
- **keep something going** (expression): to make something to continue working
e.g. This means that those elements or components have different functions or tasks inside the system and they work together to keep this system going.
- **run a program** (expression): to execute a program, to get a program going or working
e.g. Once we have this information organized and processed, we can use it to make decisions or to run programs.
- **source** (noun): where something comes from, its origin
e.g. We can use data as a source of facts that the computer processes and shapes and then, we receive this as information.
- **formula** (noun): a set of numbers, words or codes that indicates a process
e.g. When we receive data, we enter it for the computer to process it and with simple formulas, the computer can transform data into information.





Learning Points

Demonstratives as Pronouns and Adjectives

You use **demonstratives** to indicate the position of something or to give a characteristic to a noun. Although demonstrative pronouns and adjectives are similar, they have different uses.

You can find singular and plural demonstratives. We use different demonstratives to indicate how far an object is from the speaker.

	NEAR	FAR
SINGULAR	This	That
PLURAL	These	Those

For example:

*I do not understand the information in **this** file*
(the file is near the speaker)

*I see **that** application does not have the right codes.*
(the data is far from the speaker)

You can use two types of demonstratives for different purposes, so do not get confused. Let's look at each of them:

Demonstrative adjectives go before the noun and indicate how far the object is from the speaker.

For example:

This information is so valuable for the company.
That research report demonstrates the importance of systems.
These applications are a product of software engineering.
Those connections mean the data is organized.

On the other hand, **demonstrative pronouns** replace a noun that we already talked about.

For example:

*I sent you some files last night. Analyze only **those** before the meeting.*
*You talked about different features, but **these** are not clear for me, there it too much data.*



Learning Points

Examples

- *There are some basic concepts that we need to understand when talking about computers. These are information, data and systems.*
- *We can understand this term as a group of facts that are not organized.*
- *Then, we have information. This refers to processed data.*
- *When we think of a system, we have to consider two things: the elements that compose it and their functions. This means that those elements or components have different functions or tasks inside the system and they work together to keep this system going.*
- *We can talk more about that tutorial whenever you want.*

Information, Data and System Concepts

Systems concepts help us understand how systems work and how they interact. By making a clear distinction between information, data, and systems, you'll be able to interpret systems theory better.

Although there's a significant distinction between these three concepts, we often use them when we talk about systems, software engineering, and information systems.

COMPUTING AND SYSTEMS THEORY CONCEPTS	
Information	When you enter and store information in a computer, it is data. However, if you print it or process it, it becomes information.
Data	You enter it into a computer for processing, then you convert it into information with an 'easy to process' formula.
System	A group of components that work for specific objectives. For example, software and hardware integrate a computer system and work together to complete a function.